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To cite this article: Benjamin Carey (2023) *Ergodynamics of the Open Machine: Material Engagement and Modular Synthesis Performance Practice*, Contemporary Music Review, 42:3, 376-390, DOI: [10.1080/07494467.2023.2277563](https://doi.org/10.1080/07494467.2023.2277563)

To link to this article: <https://doi.org/10.1080/07494467.2023.2277563>



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Published online: 27 Nov 2023.



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Ergodynamics of the Open Machine: Material Engagement and Modular Synthesis Performance Practice

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ABSTRACT

This paper analyses some theoretical and philosophical concerns related to modular synthesis performance practice. Taking an interdisciplinary approach drawing upon concepts from cognitive archaeology, anthropology and the philosophy of technology, the paper seeks to frame modular synthesis practice as a *dance of agencies* in the context of human–machine performance. In this paper, I explore how musical ideas, technical objects and performances co-evolve as part of a set of interdependent practices. To contend with this entangled human–machine relationship, I propose a consideration of how material agency helps define the direction of these creative practices, through the process of material engagement. In doing so, we consider the unique *ergodynamics* of modular systems, and examine the interdependent relationship that exists between the two core sites of musical performance in this practice: that of ‘patching’, and of live performance. Through reflective practice and theoretical discussion, this paper analyses how an entanglement between human and material agencies challenges traditionally understood notions of musical authorship and agency.

KEYWORDS

Modular synthesis; electronic music; material engagement theory; Gilbert Simondon; ergodynamics; experimental music

Introduction

Agency and intentionality may not be properties of things; they are not properties of humans either; they are the properties of material engagement. (Malafouris 2013, 119)

As an improviser and composer, I am fascinated by the entangled relationship that exists between myself and my modular systems. In addition, I am interested in how modular systems, patches and performances *become* through this entangled process, as they are shaped by the ‘ontological force of technicity’ (Hoel and van der Tuin 2013). Through an interdisciplinary analysis that draws upon the fields of cognitive archaeology, anthropology and the philosophy of technology, this paper explores how musical ideas, technical objects and performances co-evolve as part of a set of interdependent practices. In the process, I seek to frame modular synthesis practice as a *dance of agencies* between human and material in this unique human–machine performance practice (Pickering 1995).

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Modular synthesisers are often bespoke and user-defined, and invite explorative modes of creativity (Boden 2004). In engaging with these systems, musicians form relations between complex ensembles of technical objects; building and re-defining their interactive potential from one performance to another. As prime examples of the Simondonian *open machine* (Simondon 2017), modular systems require the user to define the boundaries of their instrument for each performance, foregrounding instrument building as an inherent part of both compositional and performance practices. In this paper I propose that neither instrument building nor musical performance should be viewed as embodying a linear form of causality, that is, proceeding from human intention, through technical execution to artistic result. Instead, I argue that modular practices are characterised by a continual balance between determinism and indeterminacy, where the musician is simultaneously in tune with and surprised by complex networks of electricity. To contend with this entangled human-machine relationship, I propose a consideration of how material agency helps define the direction of these creative practices, through the process of material engagement (Malafouris 2013). In doing so, we consider the unique *ergodynamics* of modular systems (Magnusson 2019), and examine the interdependent relationship that exists between the two core sites of musical performance in this practice: that of ‘patching’, and of live performance.

Background

As an improviser and composer of electronic music, I am interested in interactivity, generativity, and the reciprocal creative dynamics that exist between my musical ideas and the machines I collaborate with to make music. These have been themes of my previous research, where I have explored the co-evolution of digital musical artefacts and their associated creative practices. Through the design and use of the *_derivations* interactive performance system, I explored the role that machine agency plays in the creative process of what I have termed the ‘performer-developer’ (Carey 2016b, 74; 2019a, 2). Designed for improvised human-machine performance, *_derivations* made use of both machine listening and context-based music information retrieval techniques to facilitate improvised encounters between instrumentalists and the computer (Carey 2016a, 161; Casey et al. 2008, 670). Through bricolage programming (McLean and Wiggins 2010, 2), this system evolved in tandem with an associated human-machine performance practice. The research examined the deeply reciprocal relationship that exists between human and machine agency in both the design and performance of such interactive systems, as I sought to understand the *dance of agency* I was experiencing as both developer of and performer with such systems (Akrich and Latour 1992; Latour 1992; Pickering 1995).

My current work has moved away from interactive computer music, and towards an intensive focus on the development of a modular synthesis practice. I have been composing and performing with modular synthesisers for the past eight years, with my practice largely centred around the development and use of a continuously evolving 12U ‘Euro-rack’ modular system (see Figure 1). In the past three years, I have been fascinated by Serge Tcherepnin’s notion of ‘patch-programmability’, a design approach that enables the low-level definition of a system’s functionality as part of the compositional process (Milano 1981). This interest has led to a series of residencies, workshops and performances at the Melbourne Electronic Sound Studio (MESS Ltd), home to the large 1975

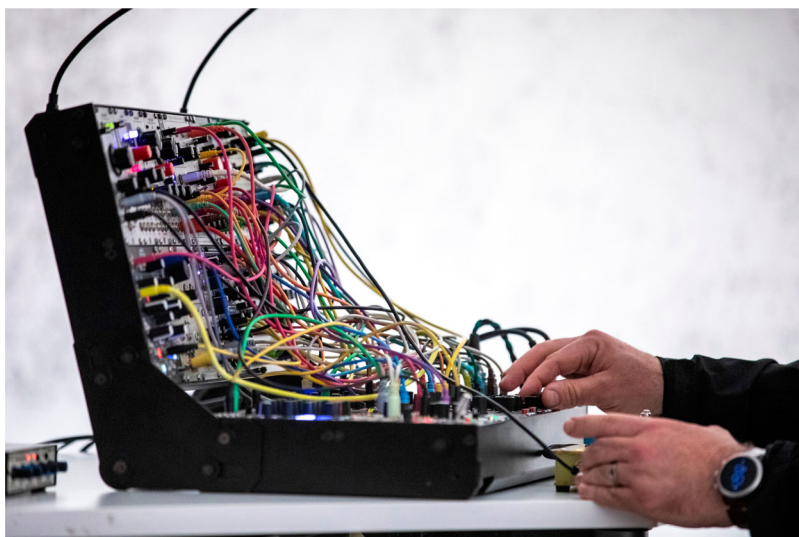


Figure 1. The author performing on a 12U Eurorack modular synthesiser at the Museum of Contemporary Art, Sydney, Australia, September, 2022 (photograph by Joseph Mayers).

La Trobe Serge 'Paperface' system (Figure 2). Most recently, I have continued these explorations through the development of my 'Kameruka' Serge system, a hybrid, 5-panel modular system based on Tcherepnin's designs.

Human–Machine Entanglement

To engage with a modular synthesiser is to both *shape* and *be shaped* by complex networks of electricity. In my work, I have noted a reciprocal relationship between myself

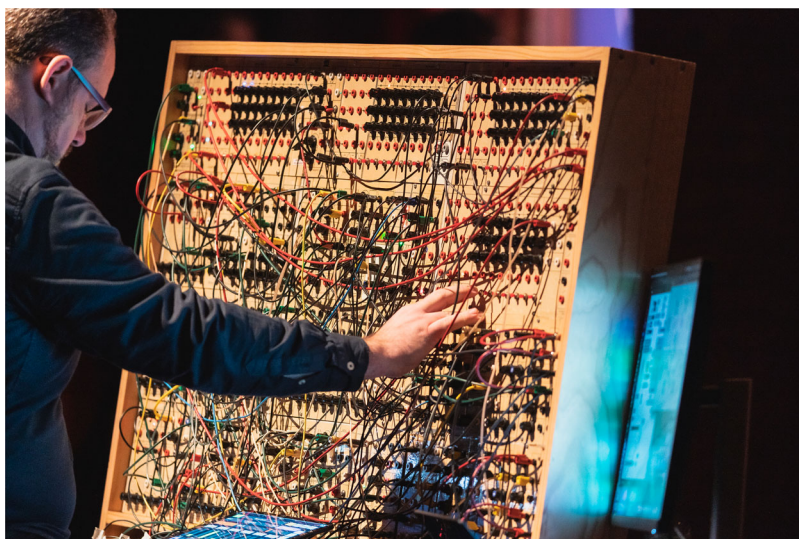


Figure 2. The author performing on the La Trobe Serge 'Paperface' modular synthesiser at The Substation, Melbourne, Australia, July, 2022 (photograph by Keelan O'Hehir).

and my modular systems, a relationship that is co-constitutive of musical ideas, technical objects and performance outcomes. This is a form of making in which *correspondence* with materials, seen as a ‘bringing forth of potentials immanent in a world of becoming’ (Ingold 2013, 31), lies at the heart of the practice.

As with other artistic practices and forms of making, modular synthesis is an engagement with the material world. In this practice, the raw materials with which the musician engages are electrical signals, represented as voltages emitted by individual *modules* present in the modular system. To engage with these voltages, the musician manipulates the panel controls of these modules, and connects modules together via patch cables in a network in the known as a *patch*. Through this process, the musician sculpts both *control voltage* and audio-rate signals into synthesised sound, designs bespoke interactive mappings and iteratively builds instruments and generative compositions. Whilst the core functionality of a module is determined by its design and manufacture, its agential properties emerge through its connection to other modules within a patch. It is within this patching process that the musician discovers and helps shape the agency exhibited by their machines. This is because modules are not simply inert technical objects, they are machines *that function*; it is what they *do*, not what they *are*, that helps determine both their utility and potential for exerting agency with the network of a modular synthesis patch.

In each engagement with a modular synthesiser, I move cyclically and performatively between designing sounds, composing musical gestures and defining performance mappings (Hunt, Wanderley, and Paradis 2002). I often begin my patching sessions with little to no pre-conceived musical outcome in mind, preferring to begin with small, local decisions, to see where they will lead me sonically and interactively. This patching approach immediately sets off a game of reciprocal causality, a cybernetic feedback loop where numerous micro-decisions are entangled with the output of the system as it grows in complexity. By being attuned to the specific sonic and interactive context of each patch as it evolves, I experience a significant push and pull between both my musical intent and the agency I sense from these systems. After many years working with these systems, I have cultivated an approach that leaves my musical intent porous, open to influence from the contingencies that emerge from my patches. Recently, I have been conceiving of my approach to performance as a kind of *structural unfurling* that takes place on stage (Carey 2019b, 4). Through the patching process, I have co-evolved both automated processes and interactive mappings with my modular system that I then reveal and interact with in performance. By surfacing and engaging with the results of complex processes, my musical decisions are made in correspondence with a cybernetic machine, the *genesis* of which sits in front of me as an imposing tangle of patch cables. My task as a performer is to unfurl and then to navigate the structure that results from the interactions between these low-level processes, revealing their inner workings and re-defining them in response to the needs of the performance as it advances forwards in time.

Given the *recursive* nature of this creative practice (Hui 2019), I argue that the musical outcomes of my engagement with modular synthesisers (including patches, live performances and fixed-media compositions) are always relational, performative and historical. To this end, they must be seen as the result of an entanglement between human and material through a *genesis* of both technicity and agential relations. To consider

agency in this way is to value the contributions of the non-human as an integral part of my practice, as equal partners in this process of ‘bringing forth’ of potentials that Ingold describes (2013, 31). In addition, this perspective acknowledges that the historicity of the practice is integral to understanding its musical manifestations. In my work, these concepts are embraced as an aesthetic position on the relationality between myself and the machines I am working with, a process that guides my musical, design, and interactive decisions at various levels throughout my practice.

Engaging and Corresponding with Materials

In his ‘Material Engagement Theory’, cognitive archaeologist Lambros Malafouris considers human cognition as deeply intertwined and co-constituted with the material world (Malafouris 2013). By demonstrating how cognition and artefacts are reciprocally developed through the process of material engagement, Malafouris rejects ‘internalist’ theories of mind that place human cognition within the brain. Instead, Malafouris argues for an enactive understanding of cognition, where human agency, intentionality and material agency are ‘brought forth’ through engagements with materials (144). In analysing the creative practice of pottery, Malafouris demonstrates the impossibility of separating cause from effect within the ‘loop of pottery making’ (Malafouris 2008, 24). In his analysis, the author reveals the inherently relational properties of intentionality in this practice, where the *sense* of agency, physical artefact and skill all emerge reciprocally through material engagement. In considering reciprocal causalities inherent in material engagement, Malafouris argues that an entanglement of agencies shapes our intentions, and in turn the artefacts we create with our materials.

Also concerned with the relational becoming of concepts and apparatuses is Andrew Pickering, a science and technology studies scholar who has theorised what he terms a *performative* view of scientific practice (Pickering 1995). Pickering’s *mangle of practice* considers how knowledge and apparatuses co-evolve in the sciences, by analysing the push and pull between humans, their apparatuses and scientific knowledge (1995). For Pickering, scientific knowledge and technical apparatuses evolve reciprocally as *accommodations* are found to *resistances* provided by the material world. This is conceived of as a *dance of agencies*, whereby human and non-human agencies push and pull on each other in a process of *interactive stabilisation*. Through a process of reciprocal causality, scientific apparatuses, concepts, and knowledge begin to stabilise (1995).

In his book *Making*, anthropologist Ingold references Pickering’s *dance of agencies*, as well as Malafouris’ theory of material engagement in outlining the difference between *interacting* and *corresponding* with materials. The author reframes Pickering’s concept as a *dance of animacy*, providing an example of kite-flying to illustrate the concept of *correspondence* as a reciprocal exchange between human and material through time, facilitated by the animacy of the air (Ingold 2013, 98–100). As both relational and temporal, correspondence is contrasted with interaction through its acknowledgement of the temporal trajectories of humans and materials, describing how engagement with material world is about moving forwards in correspondence with other things. As Ingold explains elsewhere: ‘We have been so concerned with the interaction *between* ourselves and others that we have failed to notice how both we and they *go along together* in the current of time’ (Ingold 2017, 41, italics in original).

Karen Barad, through their theory of *agential realism*, contributes a complimentary yet distinct understanding of agency and relationality with the concept of *intra-action* (Barad 2003). For Barad, the concept of agency is not a property of things or humans, it emerges through the continuous entanglement and becoming of phenomena, which they define as the ‘primary ontological unit’ (Barad 2003, 818). As contrasted with *inter-action*, the author coins the term *intra-action* to point to the continuous and relational becoming of phenomena, where things, entities and matter only exist through their *intra-activity*. As the author explains: ‘matter does not refer to a fixed substance; rather, *matter is substance in its intra-active becoming—not a thing, but a doing, a congealing of agency*’ (Barad 2003, 822, emphasis in original). From an agential realist perspective, to engage with materials is to *intra-act*, to be reciprocally entangled with phenomena through a shared becoming.

The Technicity of Material Engagement

What the above theoretical perspectives share is an acknowledgement of an emergent process of reciprocal causality that is inherent in various forms of material engagement. Whether through scientific knowledge production, artistic practice or other forms of making, these authors outline the significance of the entanglement between humans and materials in a world of becoming. By considering these ideas, we can begin to understand an approach to musical engagement with technology that ‘take(s) materials seriously’ (Ingold 2011, 31). In prosecuting their arguments, the above researchers pay particular attention to agential dances between humans and materials such as clay and stone, air and other elements. However, through a consideration of the notion of the apparatus and the artefact, we also see a concern for both the role and genesis of technology through material engagement.

Whilst a patch programmed on a modular synthesiser is no hand-axe or clay pot, I argue that these ideas have direct relevance for composers and performers working with these systems. In developing a patch on a modular synthesiser, I participate in a dance of agencies with electrical signals, as expressed by the animacy of control voltage and synthesised sound. Within the Malafourian ‘loop’ of patch-programming, cause is inseparable from effect as my patches build in complexity. By following and corresponding with materials, a process of interactive stabilisation occurs, whereby musical intentions, bespoke technical objects and musical artefacts emerge through Baradian intra-activity. By searching for possible accommodations to resistances discovered in the flow of materials, I modify my patch as it develops, exploring its boundaries and continuously updating my own perspective on its functionality. Whilst an initial intention may have driven the choice of mapping from one module to another, my compositional intent is mangled through encounters with contingent events and processes. As discussed by the philosopher of technology Yuk Hui, this is a cybernetic and organicist perspective on technological becoming, where contingencies are recursively integrated into a functioning system (2019). To understand the genesis of a modular patch is therefore to understand how these contingencies are intra-actively integrated in this practice, facilitating performative correspondences between human and machine.

Ontogenetic Practices: The Development of *Ergodynamics*

From an ontological perspective, whilst I compose and perform with these systems regularly, I do not conceive of the modular synthesiser as a musical instrument. Instead, these systems represent a field of possibilities, the *technical milieu* from which new musical instruments are evolved. Whilst the same system may be used for multiple performances, a synthesiser of this kind cannot be immediately played without first being patched, or ‘patch-programmed’. This patch-programming process is the initial site of the dance of animacy between human and machine, a place where human-material correspondences generate instruments and generative compositional structures. Much like Ingold’s ‘limp and lethargic’ kite without contact with the air (2013, 99), a collection of modules sits inert and silent without the presence of patch cables connecting modules together in a network. Each patch defines the bespoke sonic and interactive potential of the system, creating a structure that allows signals to flow between the modules, and a network of musical and interactive possibilities (Figure 3).

Considering the above, the ontology of the modular synthesiser is complicated by the fact that for each performance with the system, a new set of functionalities must be created from its constituent parts. To prepare a performance with a modular synthesiser is to first *build an instrument*, complete with its own modes of interaction, methods of sound generation and parameter mappings (Hunt, Wanderley, and Paradis 2002). The modular synthesiser may therefore be seen as a *machine for building machines*; a system assembled to facilitate the design, discovery, and use of new musical instruments. Viewed from this perspective, to perform with a modular synthesiser is to interact not with an instrument that *exists*, but with one that represents a *genesis*, an entity inseparable from its own intra-active becoming. Whilst all musical instruments have a history of design and an associated culture of use, and many are reconfigurable in the hands of the

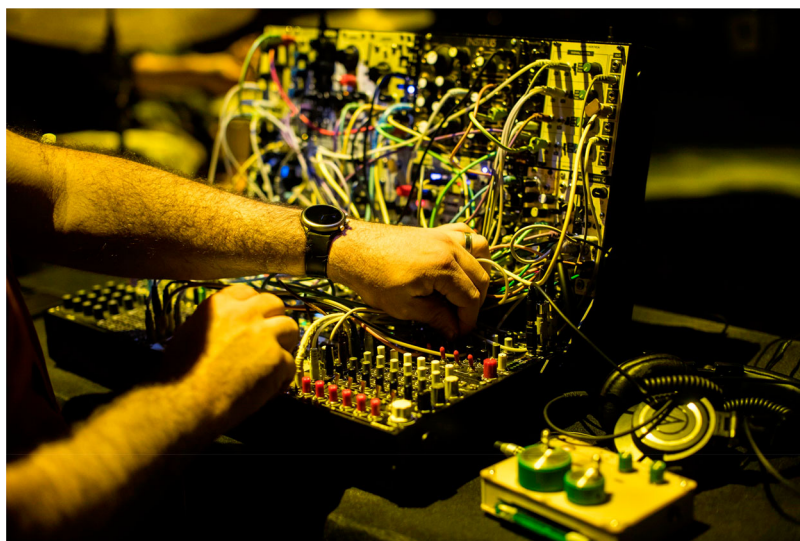


Figure 3. The author performing on a 12U Eurorack modular synthesiser at Phoenix Central Park, Sydney, April 2021 (photograph by Ken Leanfore).

exploratory musician, the modular synthesiser is a peculiar case study as each engagement with the system requires the development of functionality specific to the current interaction. It cannot make sound without being patched. Once patched, the machine carries with it the visible, audible, and interactive traces of its recent genesis.

Thor Magnusson has recently developed the term *ergodynamics* to describe the way in which a musician experiences the affordances, constraints and openness of musical instruments through cumulative engagement and play (2019). The ergodynamics of a particular musical instrument reveal themselves through interaction, with some of the most successful musical instruments enabling deep connections to be made between player and instrument over a long period of time. As Magnusson explains:

[Ergodynamics] signifies the instrument's potential for expression, what lies in it, its discoverability, mystery and magic. Ergodynamics is not necessarily about directly perceivable qualities (like affordances) or perceived limits (like constraints); it is a general concept that signifies how the instrument can be played, a historical awareness of its play, its material and virtual qualities, and what potential the instrument has for integration in wider musical contexts. (2019, 11)

When we speak of ergodynamics in the context of a modular synthesiser, we must understand this term in two separate contexts. Firstly, with respect to the ergodynamics of the system itself, and secondly, with respect to each individual patch developed on this system. As with musical instruments, modular systems possess their own ergodynamics which can be learned over time. We may call these a system's *primary ergodynamics*, the ground upon which further ergodynamics might be evolved through the patching process. Through engagement with these primary ergodynamics the musician develops an understanding of the affordances of the modules contained within their system, as well as the degree of openness they provide. That is, their potential for connectivity and interoperability. Whilst some modular systems have a fixed configuration of modules (such as the La Trobe Serge 'Paperface' (see Figure 2)), many systems are themselves part of an ongoing evolutionary process. Modules are often swapped in and out of systems, regularly changing their primary ergodynamics.

What is common in both cases, however, is that the patch-programming process represents a space in which ergodynamics themselves are intra-actively constructed, learned and explored. In my practice, whilst I have internalised the affordances of knob twisting, fader riding, joystick manipulation etc., the mapping of these physical input mechanisms is different for every performance. In my practice, the ergodynamics of the machine are still being developed and learned, and are therefore still new at the time of performance. There is hence a fresh and ongoing tension between those functions that are known and internalised, and those that force a phenomenological reckoning with their functionality.¹ Whilst evolved by the musician, the ergodynamics of a modular patch are often surprising and unexpected, full of contingent events and processes. As Magnusson observes: 'the patch is about forming the research field and setting the boundaries of thought and expression, but then it takes time and engagement to explore the ergodynamics of the patch, and get to know the system' (2019, 119).

We may thus consider the practice of patching as an intra-active activity in which ergodynamics are evolved through the recursive invention of new *open machines* (Simondon 2017). This a process of invention that involves both the design and discovery

of instrumental affordances, navigated through the process of interactive stabilisation. With specific reference to musical instruments, Magnusson's ergodynamics provides us with a relational understanding of instrumental affordances. When applied to modular synthesis practice, this concept highlights the central role that the *genesis* of technical objects plays in this practice, a concept that is central to Simondon's philosophy of technology. Expanding upon Malafouris and Pickering's analyses of artefacts and apparatuses, here I argue that modular synthesis patches may themselves be analysed as *technical objects*, bringing the practice into conversation with the ideas of the French philosopher of technology, Gilbert Simondon.

Invention, Concretisation and the Open Machine

In his work, Simondon is concerned with the relationship between humans and technology, and the importance of technological genesis in understanding the integration of technical objects in the world. By considering the genesis of technical objects, Simondon differentiates these objects from *things*, in the ontological sense. Instead, he argues for an *ontogenetic* understanding of technology, that is, that technology is a form of non-human existence inseparable from its becoming: 'a technical object is neither this thing nor that thing, given *hic et nunc*, but that of which there is genesis' (Simondon 2017, 26). Through his analyses, Simondon shows that technical objects undergo a process of *concretisation* through technical invention. Moving from technical *element*, to *individual* and finally to technical *ensembles*, technical objects move towards the concrete as they become 'pluri-functional' (Simondon 2017, xv). Whereas the technical element is an abstract object with functionality designed for a sole purpose, technical individuals are those that have been evolved to act through a process of reciprocal causality between themselves and other parts of a system: 'in it each structure exists not only as organ, but as body, as *milieu*, and as ground for other structures' (Simondon 2017, xv). In this respect, concretisation may be seen as akin to Pickering's concept of *interactive stabilisation*. Through the process of technical invention, the concretising technical object integrates contingencies into its functioning, provides accommodations to resistances encountered in its milieu, and creates an *associated milieu* for other parts of the system.

Although we may consider the modular synthesiser itself a technical object, it is in fact a collection of technical objects (modules) whose sole purpose is to be put into relation with one-another in a network. Here I argue that the patching process itself should be understood as one of technical invention, whereby new technical objects are concretised as part of the creative practice. This concretisation process becomes the site for the evolution of new ergodynamic properties to be explored by the musician. Viewed as technical elements, specific functions provided by modules in a system facilitate the evolution of new technical objects in the form of technical individuals and ensembles. Importantly, these new technical objects are specific to a particular musical intra-action. This concretisation occurs as the musician invents and evolves new methods of functioning, through the combination and recombination of elemental functions provided by the modules present in a system. To patch-programme a synthesiser is therefore to engage in an act of invention, in which a process of interactive stabilisation occurs between musician and system. This act of invention occurs as part of a recursive dance of agencies between human and material, as experienced through the primary ergodynamics of the modular system itself.

In his work, Simondon is further concerned with the concept of openness in the functioning of technical objects and machines. For Simondon, an *open machine* is one ‘endowed with a high degree of technicity’, as it possesses a certain *margin of indetermination* in its functioning (Simondon 2017, 17). Seen as a measure of openness, this margin of indetermination enables such machines to be sensitive to outside information. These are not closed systems, but those that can adapt their functioning in response to their environment, or *technical milieu*. For Simondon, what is of crucial importance for these open machines is their relationship to the humans who have both invented and continue to engage with them. In a technical ensemble, human interpretation is seen as both essential, and non-hierarchical in its relation to the regulated functioning of the technical system: ‘for there to be a true technical ensemble man has to play a functional role *between machines rather than above them*’ (Simondon 2017, 150, emphasis added). This emphasis of the in-betweenness of humans and machines is founded upon Simondon’s analysis of technical thought, where to invent is to think technically, requiring an awareness of both the historical and evolving dynamic properties of technical objects:

To invent is to make one’s thought function as a machine might function, neither according to causality, which is too fragmentary, nor according to finality, which is too unitary, but according to the dynamism of lived functioning, grasped because it is produced, accompanied in its genesis. (Simondon 2017, 150)

Discussion: *Reaching Forwards* towards Performance

The work of both Simondon and Magnusson provide further theoretical grounding for an understanding of my modular synthesiser practice as relational and intra-active. Invention, through the process of patching, is facilitated by my understanding of the ergodynamics of the synthesiser as a Simondonian *open machine*. Furthermore, I see the synthesiser as a space for the concretisation of new open machines, in the form of technical individuals and ensembles with emergent ergodynamic properties. In both the practices of patching and performance, I view my role as being between the technical objects I am evolving, as I interpret and modify their functionality in a recursive human–machine feedback loop. To invent new functionalities requires an awareness of the ongoing flow of materials in my patches, as represented by their ‘dynamism of lived functioning’ (Simondon 2017, 150). This is a process of correspondence, whereby I go along with the emergent ergodynamic properties of the machines I am intra-actively inventing (Ingold 2017, 41).

In my work, I view the patch as it exists at performance time as both performable and inherently reconfigurable, with its default states, modes of interactivity, and some of its routings left open for engagement. This is especially true when patch-programming Serge systems such as the ‘Kameruka’ or La Trobe ‘Paperface’, systems designed to facilitate low-level definition of the machine’s functionality. For me, what is of most interest is the genesis of the patches themselves, and the way in which the process of patching *reaches forwards* towards musical performance. Whilst evolving new instruments during patch-programming, I begin to congeal the interactive and sonic ideas for each part of the patch. As more sound sources and modes of interaction are added, the musical context of each existing part is altered, spurring on modifications to existing

sounds and processes. In this process, I am concretising *sub-systems* of my patches as new technical individuals, as functions and sounds developed for one purpose are put in relation to one another through their *pluri-functionality* (Simondon 2017, xv).

During this early phase, my creative process is focused on the potential of an individual sound source, and the definition of modes of interaction or control voltage automation for this source. Through a dance of agencies with modules and the electrical signals they emit, I evolve both low-level automation as well as possible modes of physical engagement with each sound source. As a global patch builds in complexity, multiple sound sources are put in resonance with each other, and a technical ensemble begins to emerge. At this point, I begin to view each source as a concretised sub-system, a technical individual that becomes intentionally black-boxed at various stages of the process (Akrich 1992; Latour 1992). As I have argued elsewhere, *intentional black-boxing* is an invaluable part of the shift from technical development to performance in performer-developer creative contexts (Carey 2016a, 200). Sound sources must be engaged with for music to be made, so it is essential for there to be a cognitive shift between instrument building and musical performance. Whilst still acting as open machines, in my work these sub-systems emerge as a series of semi-black-boxed, yet *metastable* technical individuals. As metastable, I acknowledge their potential for reconfigurability and ongoing concretisation.

In this practice, a process of instrument building reaches forwards in time towards musical performance. This process can occur in various ways, depending on the specific path through which the sound source has been evolved, that is, its *genesis*. Some sound sources may be largely designed from the outset to facilitate a particular form of control or engagement. For example, connecting the pressure output of the ‘Touch Activated Keyboard Sequencer (TKB)’ (Serge Modular 2022) to the gain input of a voltage-controlled amplifier affords tactile, physical control over the amplitude of a source. This decision, made at the outset of the concretisation process, reaches forwards towards performance by first privileging physical interaction before the sound source itself has been evolved. Whilst subsequent interactive stabilisation will congeal the sonic and interactive potential of the source, this initial decision is an integral part of its genesis, filtering through to future accommodations to resistances encountered in the patching process.

By way of contrast, another sound source might begin life as an abstract modulation idea, lacking a pre-determined interactive context. In such cases, I am interested in the evolution of complex, generative sonic entities, with multiple forms of automation defining the sound’s internal temporality. For such automated sources, their performative potential is generally left open for later concretisation. In the patch-programming process, I make relationships between the modulations driving these sources and other sub-systems in the patch. Through concretising these technical individuals, potential modes of performative engagement emerge through their intra-activity. For example, a source whose dynamics, timbre and pitch material are driven by interlocking looped function generators may be left to run freely, with the resultant sonic output simply feeding an open channel on a global mixer. Listening to this source, I might experiment with physically engaging or disengaging the cycling behaviour of a function generator, performatively starting or stopping some of these dynamic properties, playtesting this as a potential mode of performance. Further still, by interpreting its output in relation

to the technical ensemble of the global patch, I might choose to integrate this behaviour with other sub-systems, routing modulation external to this source to control this start/stop behaviour. I have now automated what was first conceived of as a performative control, concretising my technical individuals through their pluri-functionality, and leaving open future possibilities for physical control of this source.

These examples illustrate that through a process of interactive stabilisation, I learn the ergodynamics of my technical objects as I intra-actively evolve with them. The genesis of the technical ensemble of the patch is a crucial factor in how, and when, these patches begin to reach forwards towards performance. As outlined above, patch-programming is an intra-active process which continually defines and redefines the potential of the various sound sources in a patch. Given this, viewing each sub-system as a black box with defined inputs and outputs is an intentional decision, one that is in constant flux during most of the patch-programming process. From a Malafourian perspective, the intentionality behind this black-boxing process should be viewed as emergent, facilitated through a continual process of material engagement. The patch-programming process, as the site of a recursive process of interactive stabilisation, reveals both the affordances and constraints of each sub-system as they evolve over time. I am both evolving and learning bespoke ergodynamics as I go along with the voltages flowing through my system. Through a process of concretisation, I decide upon specific accommodations to the materials resistances discovered in the patching process, congealing the source into a sub-system that reaches forwards towards musical performance.

As discussed throughout this paper, the process of interactive stabilisation is not purely technical in this practice, but also cognitive. Whilst the black boxes that develop through patch-programming are not physically closed, I conceive of these sub-systems as metastable because they are defined as much by their physical routings as by my understanding of their appropriate methods for engagement. Modular synthesis is a creative process whereby concepts and musical intent are evolved alongside the technical objects used to perform and record new music. As concretising technical individuals, sub-systems begin to congeal as they are put in resonance with each other, with this stabilisation process being driven by my perception of their musical potential in the context of the global patch. By intentionally black-boxing these sub-systems at various points in the process, they have become ‘instruments of knowledge’ (Akrich 1992, 221), facilitating creative interpretation of their functionality as I *go along* with them in a process of performative correspondence (Ingold 2017, 41).

Conclusion

Throughout this paper, I have sought to explore the complexities of my modular synthesis practice by engaging with theoretical concepts from an interdisciplinary array of fields, including cognitive archaeology, anthropology, science and technology studies and the philosophy of technology. I have been drawn to this diverse body of research through an intuition about the co-constituted nature of musical and technical invention in my practice. Machine agency, human intentionality and a sense of shared human-machine authorship are common themes in my research. In working with these systems, I have become fascinated by how my musical decisions are made in collaboration with the machines I engage with as part of my practice.

For Malafouris, human cognition is inseparable from corporeal engagements with the world, and human intentionality is formed through the process of material engagement. Through patch-programming modular synthesisers, my musical intentions may be said to emerge along with the material agency exerted by newly evolved technical objects. Pickering's *dance of agencies* shows that knowledge, agency and apparatuses form and reform through a process of *interactive stabilisation*. By seeking *accommodations* to material *resistances* discovered in practice, I argue that I engage in a process of interactive stabilisation of my technical objects, concepts, and musical works. Through a consideration of Ingold's concept of *correspondence*, we have considered how both human and materials *go along together*, influencing each other reciprocally as they move forward in time. In considering the historical aspect of modular synthesis practice, we considered the work of Barad and Simondon, both of whom are concerned with the *ontogenetic* qualities of phenomena and technicity. By reading modular synthesis practice through a Simondonian lens, we conceived of patching as a process of technical *invention*, whereby new ergodynamics are evolved and explored as an integral part of the creative practice. By being *between* my machines as they evolve, I assist them in reaching forwards towards performance, through the *intra-active* invention and *concretisation* of technical *individuals* and *ensembles*.

Note

1. In Magnusson's work, this aspect of ergodynamics is analysed with respect to the Heideggerian concepts of the 'ready-at-hand' and 'present-at-hand' (2019, 170). See also (Hamman 1999) for a related discussion in the context of computer music practice.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Funding

This work has been supported by the University of Sydney as part of the 2022/23 SOAR Prize, a two-year internally funded programme of research awarded to Early and Mid-Career Researchers at the University of Sydney.

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